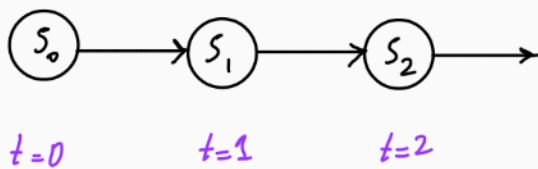


Graphical Representation of MC and MDP

Temporal Unrolling

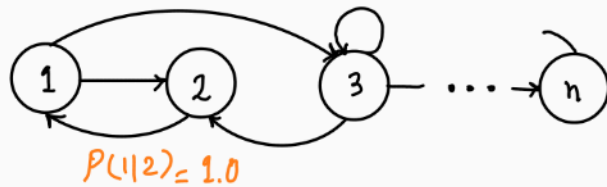
MC



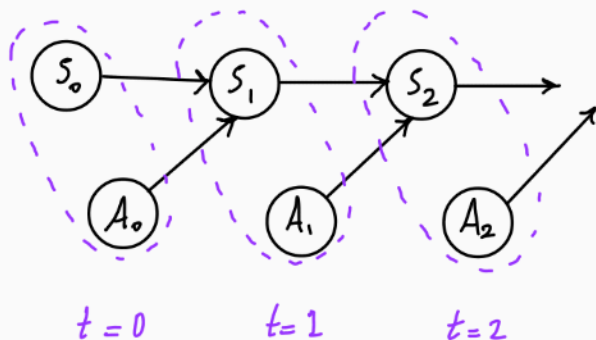
Transition - Based

$$P(3|1) = 0.4$$

$$S = \{1, 2, \dots, n\}$$

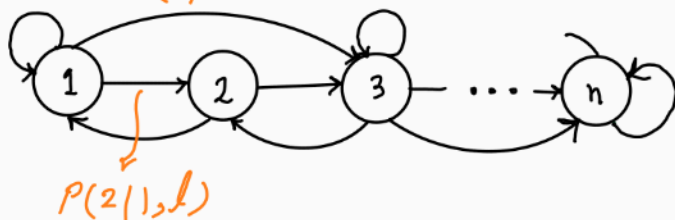


MDP



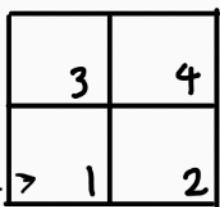
$$P(1|1, r)$$

$$P(3|1, r)$$



MDP + policy $\rightarrow$ (induced) MC

edge label: (action, transition probability)



$$S = \{1, 2, 3, 4\}$$

$$A = \{r, l, u, d, s\}$$



Policy

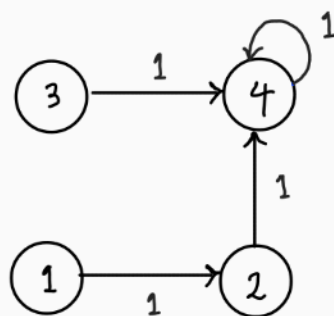
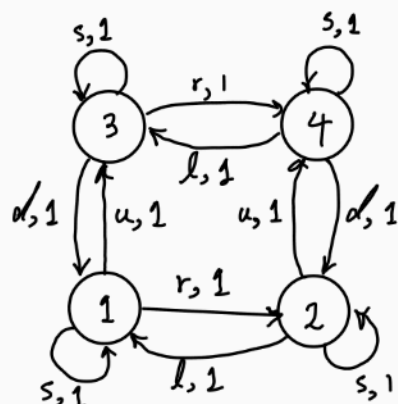
- if $s = 1 \rightarrow a = r$
- if $s = 2 \rightarrow a = u$
- if $s = 3 \rightarrow a = r$
- if $s = 4 \rightarrow a = s$

(induced)

MDP



MC



state Distribution over MC

$$\mu_t \in \Delta(S)$$

$$\mu_t(s) = P[S_t = s]$$

$$S = \{0, 1\}$$

$$\Delta(S)$$

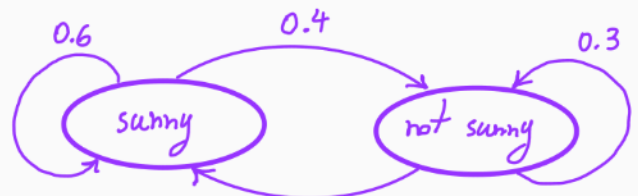
$$\begin{matrix} \xrightarrow{s=0} & \xrightarrow{s=1} \\ \downarrow & \downarrow \\ \mu = [0.4 & 0.6] \end{matrix}$$

$$\mu_{t+1}(s') = P[S_{t+1} = s']$$

$$= \sum_{s \in S} P[S_{t+1} = s', S_t = s]$$

$$= \sum_{s \in S} P[S_t = s] P[S_{t+1} = s' | S_t = s]$$

$$= \sum_{s \in S} \mu_t(s) P(s' | s)$$



$$\begin{cases} \mu_{10}(\text{sunny}) = 1 \\ \mu_{10}(\text{not sunny}) = 0 \end{cases}$$

$$\begin{cases} \mu_{11}(\text{sunny}) = 0.6 \\ \mu_{11}(\text{not sunny}) = 0.4 \end{cases}$$

$$\begin{cases} \mu_{12}(\text{sunny}) = 0.6 \times 0.6 + 0.4 \times 0.7 \\ \mu_{12}(\text{not sunny}) = 1 - \mu_{12}(\text{sunny}) \end{cases}$$

row vector

$$\text{Let } \mu_t = [P[S_t=1] \quad P[S_t=2] \quad \dots \quad P[S_t=n]]$$

$$P = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{n \times n}$$

$P(j|i)$

$$\mu_{t_0+1} = \mu_{t_0} P$$

$$\mu_{t_0+2} = \mu_{t_0+1} P = \mu_{t_0} P^2$$

⋮

$$\mu_{t_0+t} = \mu_{t_0+t-1} P = \mu_{t_0+t-2} P^2 = \dots = \mu_{t_0} P^t$$

$$\begin{bmatrix} \square \end{bmatrix}_{\mu_{t+1}} = \begin{bmatrix} \text{---} \end{bmatrix}_{\mu_t} \begin{bmatrix} \text{---} \end{bmatrix}_P$$

Chapman-Kolmogorov Equation

column vector

$$\mu_t = \begin{bmatrix} P[S_t=1] \\ P[S_t=2] \\ P[S_t=n] \end{bmatrix}$$

t-step transition matrix

$$\mu_{t_0+1} = P^T \mu_{t_0} \longrightarrow \mu_{t_0+t} = (P^t)^T \mu_{t_0}$$

Stationary (steady state) distribution: A state distribution $\bar{\mu}$

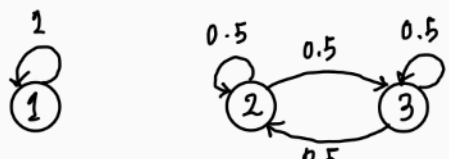
such that
$$\begin{cases} \bar{\mu}(s) \geq 0 & \forall s \in S, \quad \sum_{s \in S} \bar{\mu}(s) = 1 \\ \bar{\mu} = \bar{\mu} P \end{cases}$$

left eigenvector corresponding to eigenvalue 1

$$[\bar{\mu} P^2 = \bar{\mu} P P = \bar{\mu} P = \bar{\mu} \quad \dots \rightarrow \bar{\mu} P^t = \bar{\mu} \quad \forall t \in \mathbb{N}_0]$$

Example:

not irreducible 1)

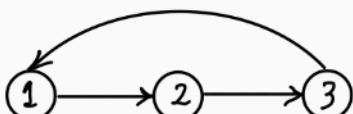


$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

$$\bar{\mu}^1 = [1 \ 0 \ 0]$$

$$\bar{\mu}^2 = [0 \ 0.5 \ 0.5]$$

periodic 2)



$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\bar{\mu} = [\frac{1}{3} \ \frac{1}{3} \ \frac{1}{3}]$$

if $\mu_t = [0 \ 0.7 \ 0.3] \rightarrow \mu_{t+1} = [0.3 \ 0 \ 0.7] \rightarrow \mu_{t+2} = [0.7 \ 0.3 \ 0]$
 $\rightarrow \mu_{t+3} = [0 \ 0.7 \ 0.3] \quad \dots$

Theorem: An ergodic (irreducible and aperiodic) has a unique stationary distribution $\bar{\mu}$.

Perron -
Frobenius
Theorem

$$\lim_{t \rightarrow \infty} P^t = \mathbf{1} \bar{\mu}$$

$$\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} [\bar{\mu}(1) \ \bar{\mu}(2) \ \dots \ \bar{\mu}(n)] = \begin{bmatrix} \bar{\mu}(1) & \bar{\mu}(2) & \dots & \bar{\mu}(n) \\ \hline \vdots \end{bmatrix}$$

Note: The speed of convergence (starting from any $\mu_0 \in \Delta(S)$) to the stationary distribution $\bar{\mu}$ is characterized by $\frac{\lambda_2}{\lambda_1}$.

$$1 = \lambda_1 > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$$

$$|\frac{\lambda_2}{\lambda_1}| \downarrow \rightarrow \text{speed of convergence} \uparrow$$

Example:

$$S = \begin{bmatrix} \text{sunny} & \text{not sunny} \\ s & n \end{bmatrix}$$

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.7 & 0.3 \end{bmatrix} \begin{matrix} s \\ n \end{matrix}$$

$$\mu_0 = [\mu_0(s) \quad \mu_0(n)] \in \Delta(S)$$

$$\bar{\mu} = \bar{\mu}P \rightarrow \bar{\mu}(I - P) = \mathbf{0}_{1 \times n} \rightarrow \bar{\mu} = \begin{bmatrix} \frac{7}{11} & \frac{4}{11} \end{bmatrix}$$

$$\mu_0 \rightarrow \mu_1 \rightarrow \mu_2 \rightarrow \dots \xrightarrow{\text{as } t \rightarrow \infty} \bar{\mu}$$

